

Servo is falling behind the web. It completes **~22 features/year** at production quality, while the Baseline “**Widely Available**” set grows at **~52 features/year**. At this pace, Servo plateaus around **80% by ~2037 and never catches up**. Full velocity parity requires **~44 FTE (€8.8M/yr, €26.3M over 3yr)**. BWA features vary widely in real-world usage — a phased approach that prioritizes high-usage features first could reduce the initial headcount and cost by up to 40% while building toward full parity. The highest-leverage actions are **unstalling 141 blocked features** and fixing **51 regressions** (features that lost >5 percentage points). This is a work-in-progress analysis focusing on one aspect of web engine development — feature-level readiness — and is not exhaustive. It is designed to orient thinking about cost and timelines. The central question: **what would it cost in engineering investment alone to bring Servo to velocity parity within 3 years**, excluding non-engineering and operational costs?

CURRENT BWA READINESS

19.8%

87 of 439 measurable at ≥95%

BWA GROWTH

593 → ~879

today → by 2031 (~52/yr)

WEB FEATURE VELOCITY GAP

22 vs 52

Servo completes 22 features/yr
Web adds 52 new features/yr

VELOCITY PARITY

44 FTE

€8.8M/yr — €26.3M over 3yr
(13 existing + 31 new)

USAGE-PRIORITIZED PARITY

38–41 FTE

€7.6–8.2M/yr
skip low-usage (<1–5%) features

INVESTMENT SCENARIOS (MOVING TARGET)

FTE	Compl/yr	vs All BWA (52/yr)	vs <1% (49/yr)	vs <5% (47/yr)	€/yr	3yr
13 (now)	~22	✗	✗	✗	€2.6M	€7.8M
25	~35	✗	✗	✗	€5.0M	€15.0M
38	~47	✗	≈	✓	€7.6M	€22.7M
41	~49	✗	✓	✓	€8.2M	€24.5M
44	~52	✓ parity	✓	✓	€8.8M	€26.3M
55	~60	✓	✓	✓	€11.0M	€32.9M

All BWA = 593 features (~52/yr). <1% = drop 35 features below 1% usage (~49/yr). <5% = drop 56 features (~47/yr). Completions/yr = 22 × (FTE/13)^{0.7}. Cost = FTE × €200k (€150k base × 1.33× specialization). Brooks's Law scaling exponent 0.7.

FEATURE READINESS TODAY

■ 87 full (19.8%) ■ 333 partial (75.9%) ■ 19 unsupported (4.3%)

Of 593 BWA features, 439 have WPT test coverage. 154 lack WPT mapping: 78 JS built-ins (SpiderMonkey/test262), 23 semantic HTML elements, 53 other (WebGL extensions, DOM, CSS, etc.). The 78 JS built-ins and 23 semantic HTML elements (~66% of unmeasured) are supported via SpiderMonkey and the HTML parser respectively; the remaining 53 have unknown status. WPT pass rate doubled from 30% to 62% over 2.5 years.

BWA GROWTH PIPELINE

Year	New BWA	Cumulative	Source
2026	+61	654	Known pipeline
2027	+42	696	Known pipeline
2028	+27	723	Known pipeline
2029+	~52/yr	~775+	Projected avg

KEY STRATEGIC FACTORS

- 1 **Velocity gap** — Servo completes 22 features/yr vs 52 new BWA/yr. At 13 FTE it plateaus at ~80% by ~2037 and never catches up. This is the core funding case.
- 2 **141 stalled features** have zero velocity. These need architectural work and block milestones above 50%.
- 3 **51 regressions** — features that lost >5 percentage points. Fixing these is the cheapest path to closing the gap.
- 4 **154 unmeasured features** — 78 JS built-ins and 23 semantic HTML elements are supported via SpiderMonkey and the HTML parser; 20 WebGL extensions depend on GPU drivers; 33 have unknown status. Closing the measurement gap could lift readiness without engine work.
- 5 **Strategic focus > headcount**. Doubling FTE cuts time ~40% due to Brooks's Law. Targeted investment in regressions and stalled features delivers more progress per euro than broad hiring.

METHODOLOGY

Readiness measured against Baseline “Widely Available” (BWA) web features — supported 30+ months across all major browsers. Feature scores from Web Platform Tests (WPT) via WPT Feature Manifest mapping. 439 of 593 BWA features are measurable; 154 lack WPT mapping (78 JS built-ins tested by test262 not WPT, 23 semantic HTML elements, 53 other with unknown status). FTE-equivalent from per-author commit frequency (1 FTE ≈ 22 commits/month, capped at 1.0 per person; 115 authors sum to ~13 FTE). BWA growth rate of ~52/year is the 2023–2025 average, normalized to exclude the 2022 interop spike. Known pipeline: 130 features graduating by 2028. Velocity parity = Servo’s annual feature completion rate ≥ BWA annual growth rate. Completions/yr = 22 × (FTE/13)^{0.7} (sublinear Brooks’s Law scaling). Stalled = zero or negative velocity over the full observation period (2023-Q3 to 2026-Q1). Regressions = features that lost >5 percentage points. Cost at €200k/yr per FTE: €150k European senior SWE median total cost × 1.33× premium for browser-engine specialization (small talent pool, Rust/rendering/layout) and multi-disciplinary practice (software development + W3C/WHATWG standards participation + open-source community management). Calibrated against NLnet (€117k, up to €65/hr), Sovereign Tech Fund (€79–101k, TVöD-Bund + employer costs), Mozilla Germany (€145–164k), and Servo grant rates (€248k US contractor). Excludes other opex, infrastructure, management overhead, and inflation. Usage-prioritized scenarios: at <1% threshold, 35 features deprioritized (target 558, growth ~49/yr); at <5%, 56 features deprioritized (target 537, growth ~47/yr). Data: wpt.fyi, web-features, Servo git history. Full interactive dashboard: dashboard.html.

Thanks to the people who’ve reviewed the methodology and approach so far. NOTE: AI was used in the making of this draft report. For suggestions for improvement or to report inaccuracies, contact dietrich@webtransitions.org.